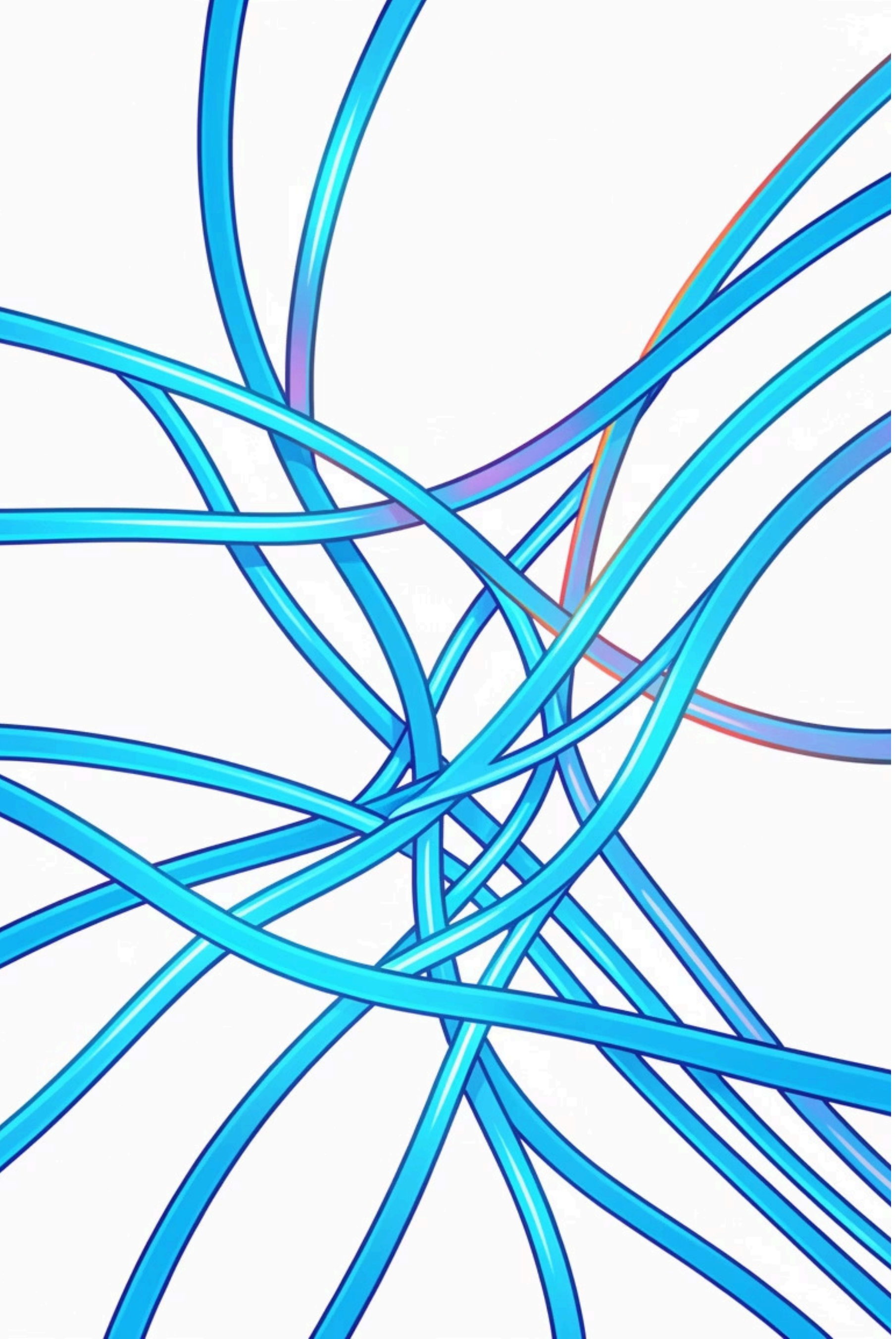


Computer Networks Course Overview

A comprehensive guide to mastering fundamental networking concepts, protocols, and architectures for undergraduate computer science and engineering students.

COURSE ROADMAP

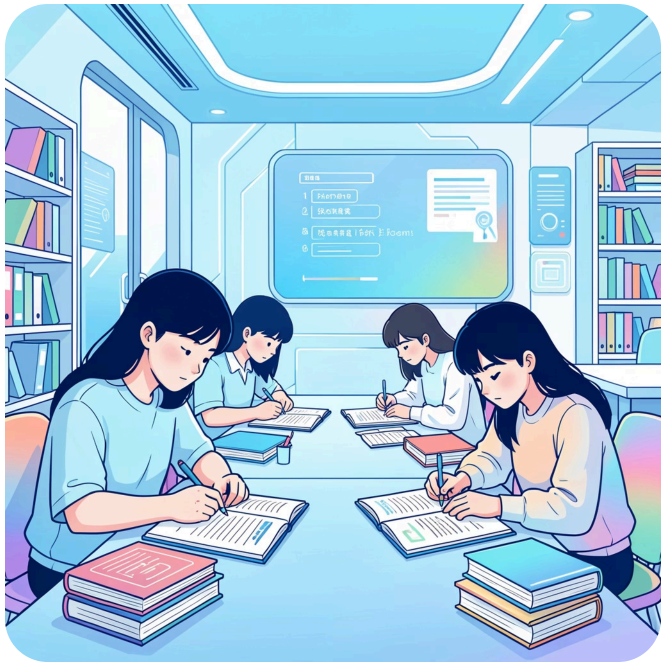
Journey Through Network Fundamentals

Instructor : Kshitij Sharma

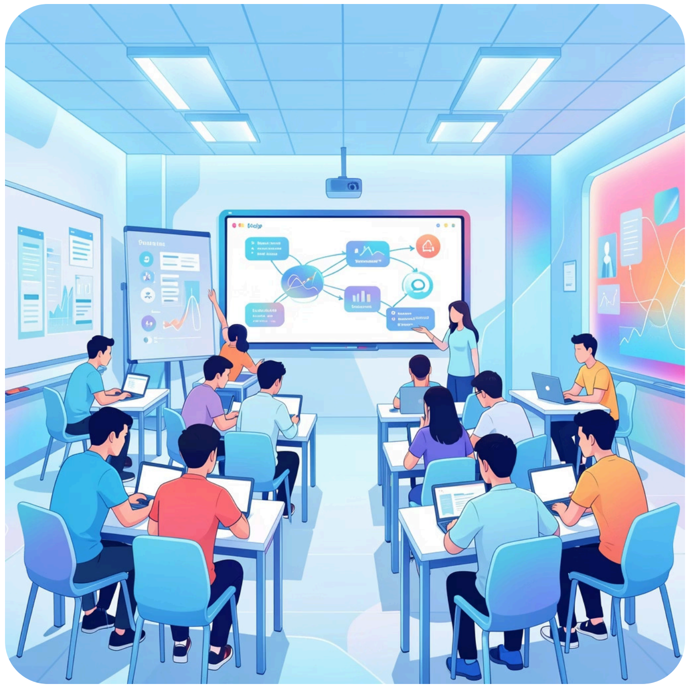
This course covers the complete networking stack—from physical media access control to application layer protocols.

We'll explore how data travels across networks, examining protocols, error handling, routing mechanisms, and security considerations that make modern internet communication possible.

Course Is Meant For:



GATE Aspirants



University Students

Interview Preparation

Our Methodology:

1

Theory

2

Practice

3

DPP

4

Revision

5

Certificate

IPv4 Addressing Architecture

Foundational Concepts

- IP addressing fundamentals and structure
- Classful addressing (A, B, C, D, E)
- Unicast, multicast, and broadcast communication
- Subnet masks and network segmentation

Advanced Techniques

- Classless Inter-Domain Routing (CIDR)
- Variable Length Subnet Masking (VLSM)
- Supernetting for route aggregation
- Efficient IP address allocation strategies

Error Detection and Correction

Networks introduce errors during transmission. Understanding detection and correction mechanisms ensures reliable data delivery across unreliable channels.

Simple Parity

Single bit appended to detect odd number of bit errors in transmitted data frames.

2D Parity

Matrix-based approach detecting errors in rows and columns for better accuracy.

Checksum

Arithmetic sum verification used primarily in TCP/UDP for segment integrity.

CRC

Polynomial division method offering robust error detection for data link layer.

Hamming Code

Error correction technique that identifies and fixes single-bit errors automatically.

Exam Focus: Cyclic Redundancy Check (CRC) is particularly important—master polynomial division and syndrome calculation.

Flow Control Mechanisms

1

Stop & Wait

Sender transmits one frame, waits for acknowledgment. Simple but inefficient for high-latency links.

2

Go-Back-N

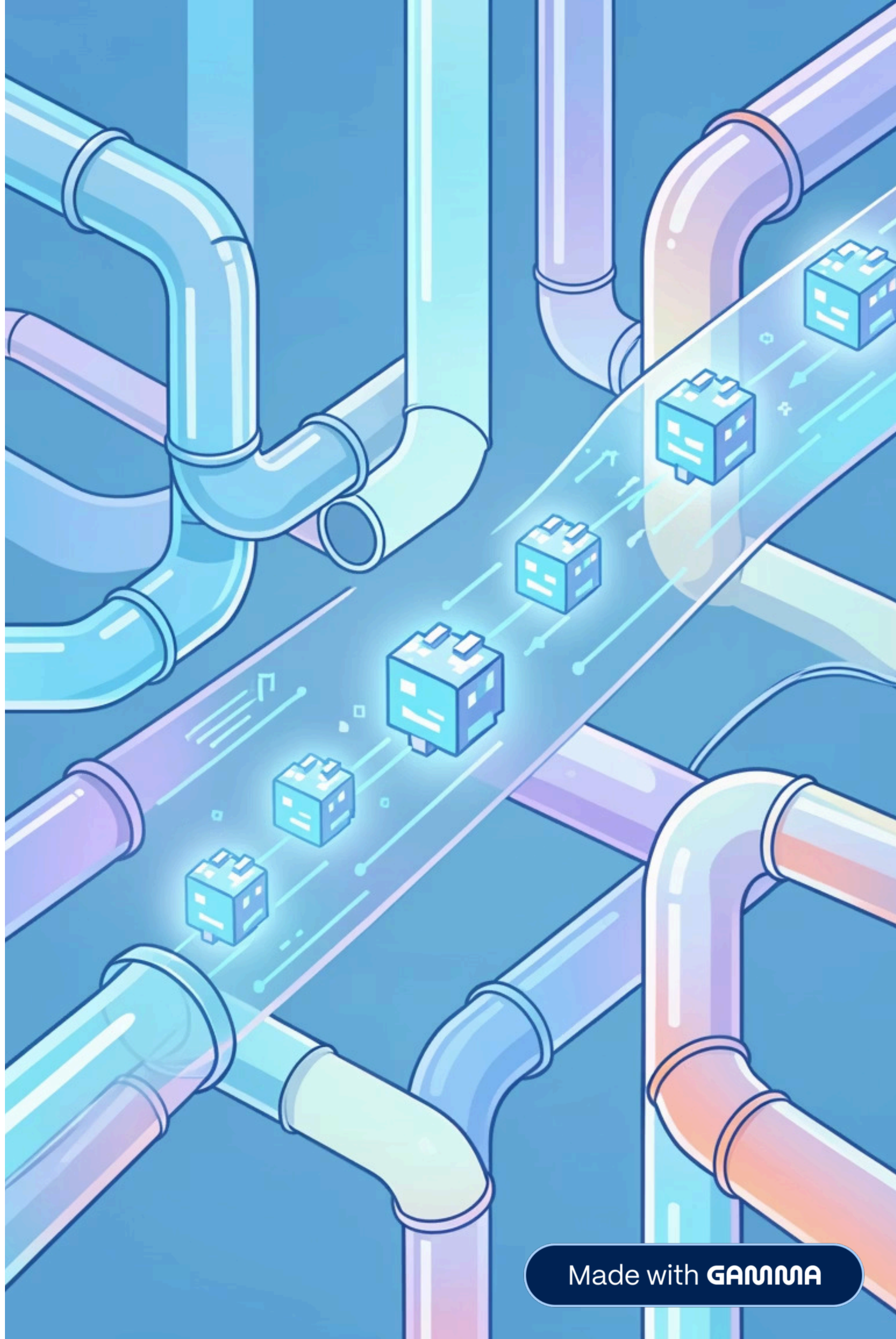
Sliding window protocol allowing multiple unacknowledged frames. Retransmits all frames from error point.

3

Selective Repeat

Most efficient approach—retransmits only corrupted or lost frames, maximizing throughput.

Understanding propagation, transmission, queuing, and processing delays helps optimize protocol selection for different network conditions.



Transport Layer Protocols

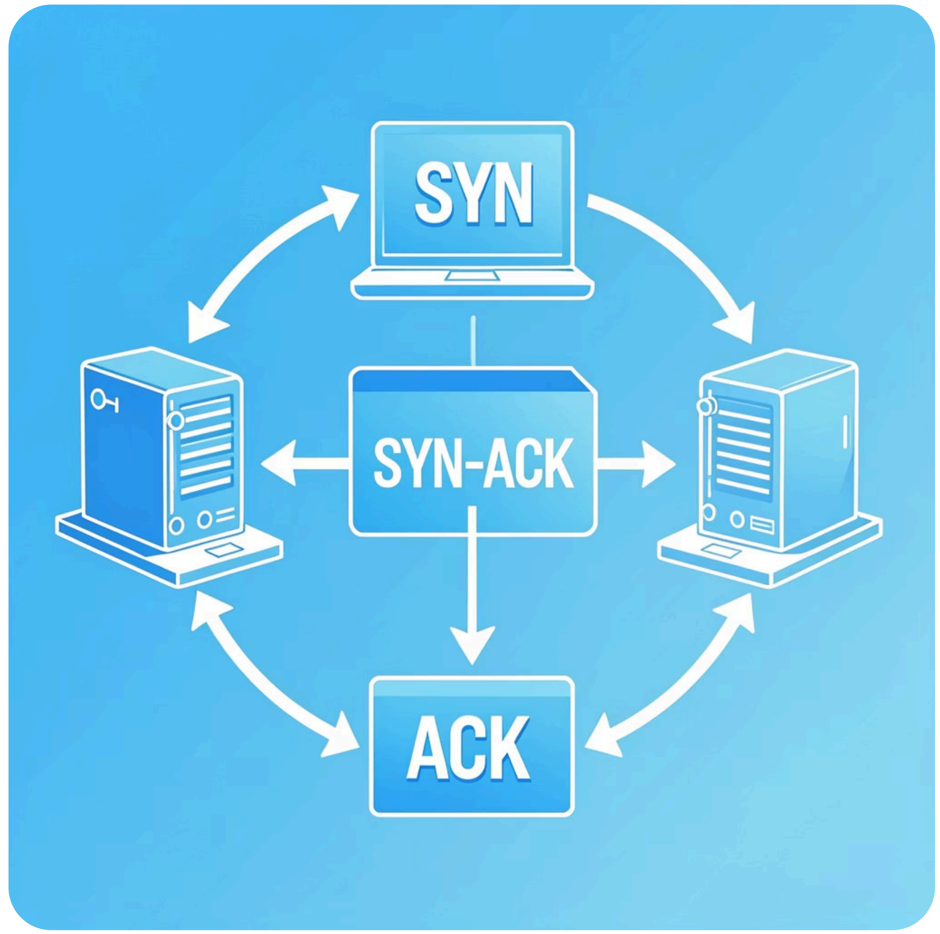
TCP: Transmission Control Protocol

- **Connection-oriented:** Three-way handshake (SYN, SYN-ACK, ACK)
- **Reliable delivery:** Sequence numbers, acknowledgments, retransmission
- **Flow control:** Sliding window prevents receiver overflow
- **Congestion control:** Slow start, congestion avoidance, fast retransmit/recovery
- **State management:** LISTEN, ESTABLISHED, TIME_WAIT, CLOSED

Critical Topic: TCP congestion control algorithms are frequently tested—understand AIMD, slow start threshold, and timeout calculations.

UDP: User Datagram Protocol

- **Connectionless:** No handshake or state maintenance
- **Best-effort delivery:** No guarantees on order or arrival
- **Low overhead:** Minimal 8-byte header
- **Use cases:** DNS queries, streaming media, VoIP, gaming



Media Access Control Protocols

Multiple devices sharing transmission media require coordination mechanisms to avoid collisions and maximize channel utilization.

Random Access

ALOHA: Pure and slotted variants for contention-based transmission

CSMA/CD: Carrier sensing with collision detection (Ethernet)

CSMA/CA: Collision avoidance for wireless networks (Wi-Fi)

Controlled Access

Reservation: Time slots allocated through reservation requests

Polling: Central controller queries each station sequentially

Token Passing: Special frame grants transmission permission

Channelization

FDMA: Frequency division for simultaneous transmissions

TDMA: Time slot allocation for multiple users

CDMA: Code division allowing overlapping transmissions

Exam Focus: ALOHA efficiency calculations, CSMA/CD slot time, and FDMA bandwidth allocation are high-yield topics.

Routing and Switching Fundamentals

Routing Protocols

Distance Vector: Bellman-Ford algorithm, periodic updates, count-to-infinity problem. Examples: RIP.

Link State: Dijkstra's algorithm, topology flooding, faster convergence. Examples: OSPF.

Other approaches: Shortest path algorithms, flooding for robustness.

Switching Techniques

Circuit Switching: Dedicated path for entire session (traditional telephony).

Packet Switching: Data divided into packets, independent routing, statistical multiplexing.

Virtual Circuits vs Datagrams: Connection-oriented versus connectionless packet delivery.

Distance Vector Routing is marked as important—understand routing table updates, split horizon, and poison reverse.

Application and Support Protocols

DNS

Domain Name System translates human-readable names to IP addresses using hierarchical distributed database.

SMTP & Email

Simple Mail Transfer Protocol handles message transfer between mail servers with MIME extensions.

FTP

File Transfer Protocol enables reliable file uploads/downloads using separate control and data connections.

HTTP

Hypertext Transfer Protocol powers the web with request-response model and stateless architecture.

ARP/DHCP

Address Resolution Protocol maps IP to MAC. DHCP automates IP configuration and address assignment.

ICMP

Internet Control Message Protocol provides error reporting and diagnostic capabilities (ping, traceroute).

Essential Resources for Success

Primary Textbook

Data Communications and Networking by Behrouz A. Forouzan

Comprehensive coverage with clear explanations, excellent diagrams, and practice problems aligned with course topics.

Supplementary Reference

Computer Networks by Andrew S. Tanenbaum

In-depth theoretical foundation with historical context and detailed protocol analysis for advanced understanding.

Alternative Perspective

Computer Networking: A Top-Down Approach by Kurose and Ross

Application-layer-first approach with real-world examples and programming assignments for practical insight.

Study Strategy: Focus on topics marked as "Important" (CRC, ALOHA, CSMA/CD, FDMA, Distance Vector, Packet Switching, TCP Congestion Control). Practice numerical problems for subnetting, error detection, and efficiency calculations. Review protocol state diagrams and header formats thoroughly.